CSME-CFDSC-CSR
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Characteristics of street canyon flow using Large-Eddy Simulation with a drag-porosity modelled atmospheric boundary layer

Authors:

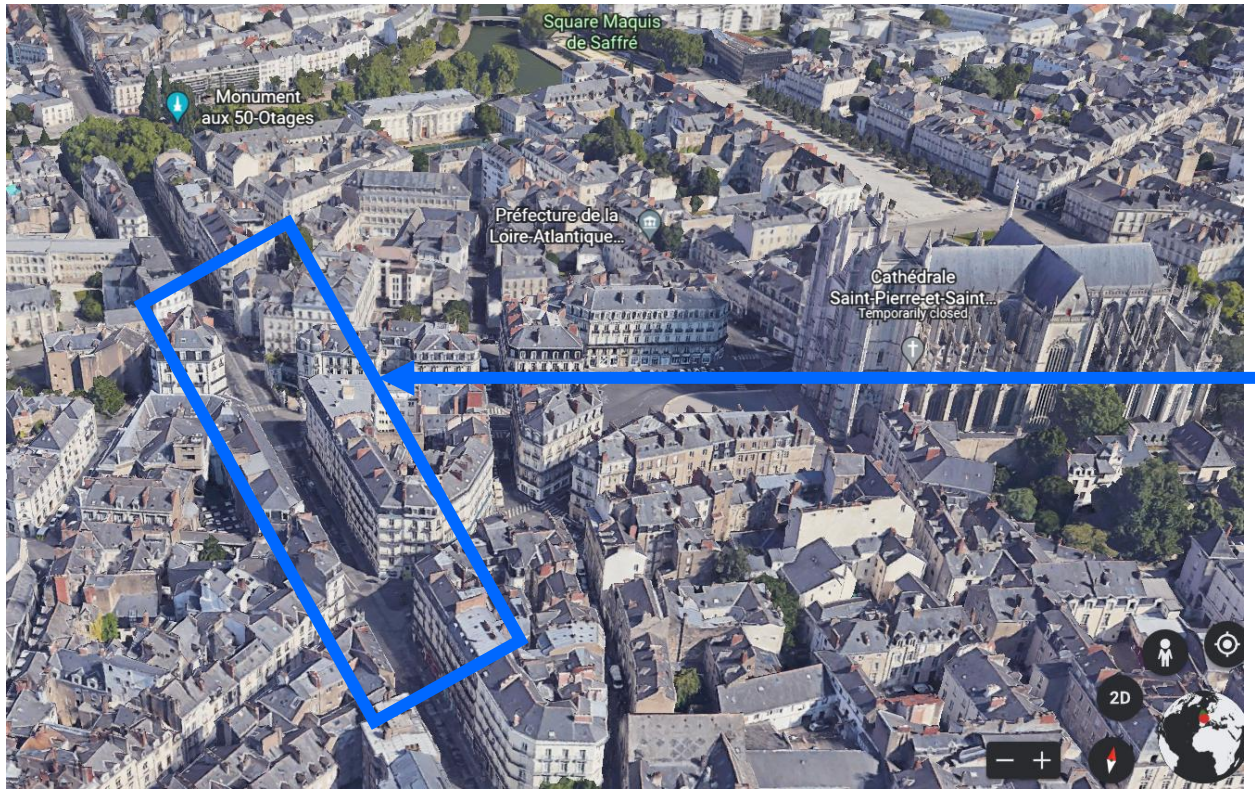
Haoran Du 

Laurent Perret 

Eric Savory 

Background

> Background



Real Satellite Google Map-Rue de Strasbourg (RdS)

Poor urban air quality

Dynamic transport of air and pollutants

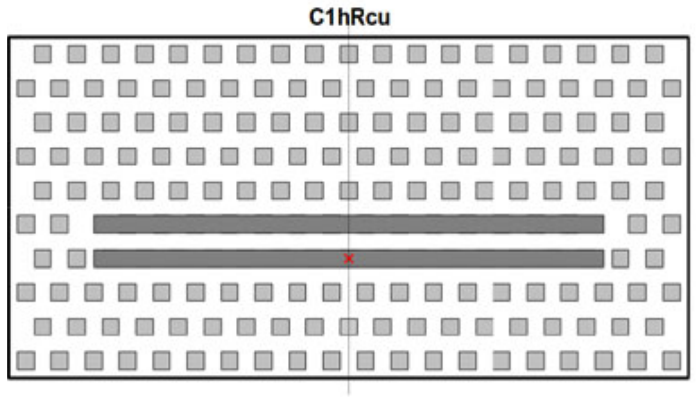
Street canyon model 2

Urban boundary layer 1

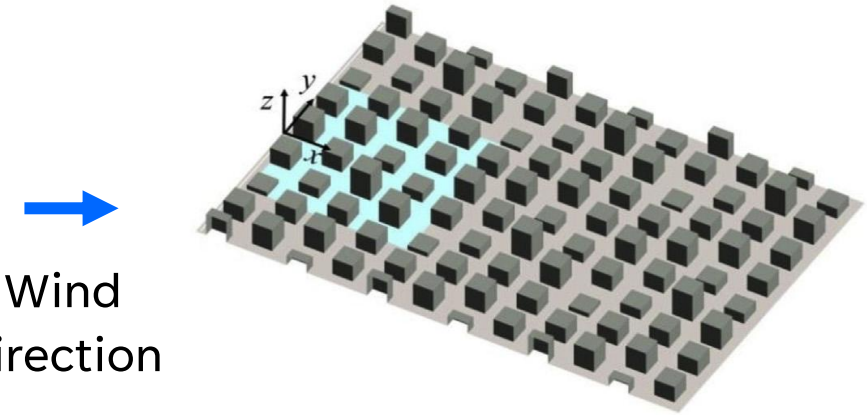
Interaction

GOAL

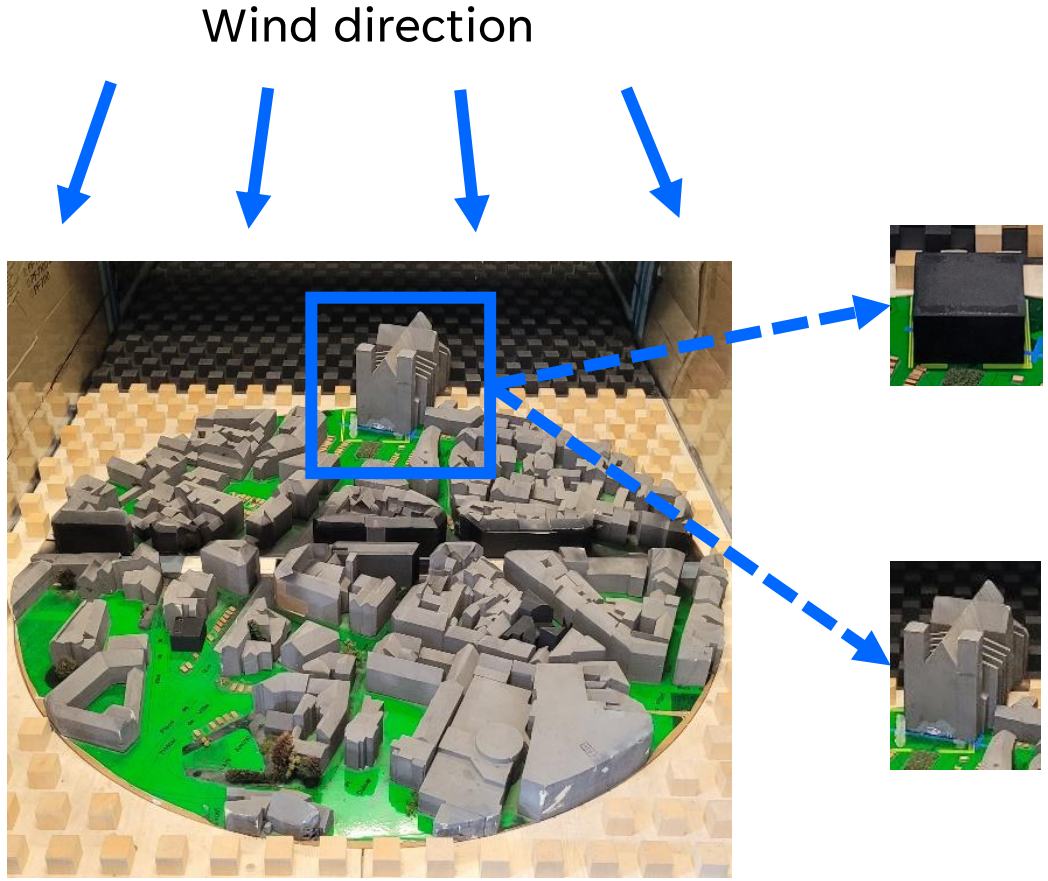
Urban-scale dynamical model



Homogeneous model
(Blackman et al., 2015)



Heterogeneous model
(Goulart et al., 2019)



Morphological model
(Du et al., 2023 & 2024)

Papers from the 80s already introduced drag porosity method for vegetation canopy flow.

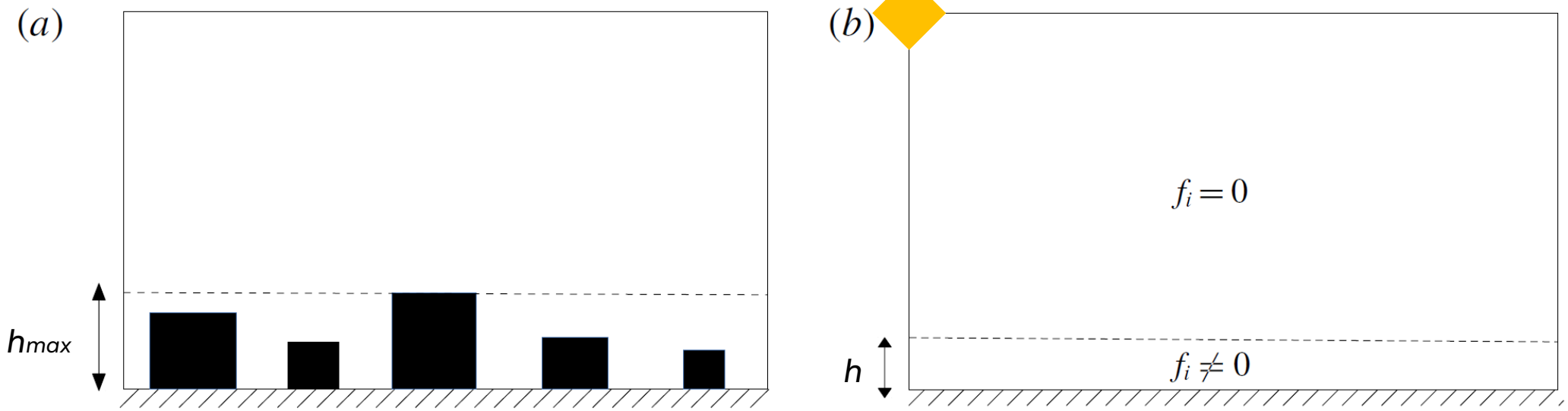
(Tian, 2018)	(Varghese and Durbin, 2020)

"A corollary to Townsend's hypothesis: roughness can be replaced by a force that produces the same drag as the geometrical roughness. "

"Large scale turbulence structures are captured, as well as the Reynolds stresses."

$$\frac{\partial \hat{u}_i}{\partial t} + \hat{u}_j \frac{\partial \hat{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \hat{p}}{\partial x_j} + \frac{\partial}{\partial x_j}\left(\nu \frac{\partial \hat{u}_i}{\partial x_j} - \tau_{ij}^{SGS}\right) - f_i$$

Additional momentum sink



$$\frac{\partial \hat{u}_i}{\partial t} + \hat{u}_j \frac{\partial \hat{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \hat{p}}{\partial x_j} + \frac{\partial}{\partial x_j}\left(\nu \frac{\partial \hat{u}_i}{\partial x_j} - \tau_{ij}^{SGS}\right) - f_i$$

Roughness frontal area per fluid volume

$$\frac{1}{2} C_d a_f \tilde{u}_i \sqrt{\tilde{u}_i \tilde{u}_j}$$

Drag coefficient

WHY?

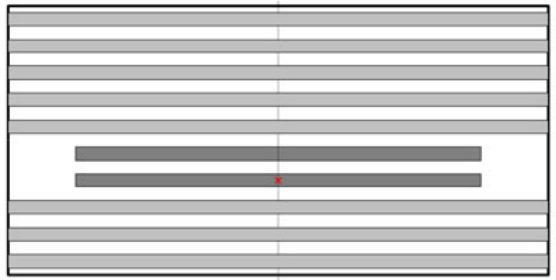
- **Proven to be successful in ARPS*** (Bucquet, 2023)
- **It does not require very fine meshes within the drag porosity layer.**

- **Precursor simulations with periodic boundary conditions**
- **Stability at certain locations, i.e. outlet**

* - The Advanced Regional Prediction System (ARPS)

Wind direction

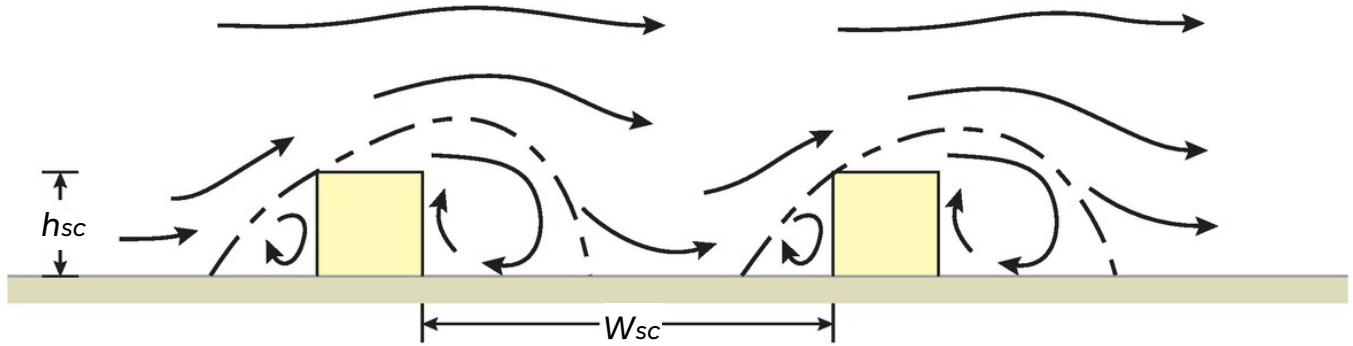
Look from the side



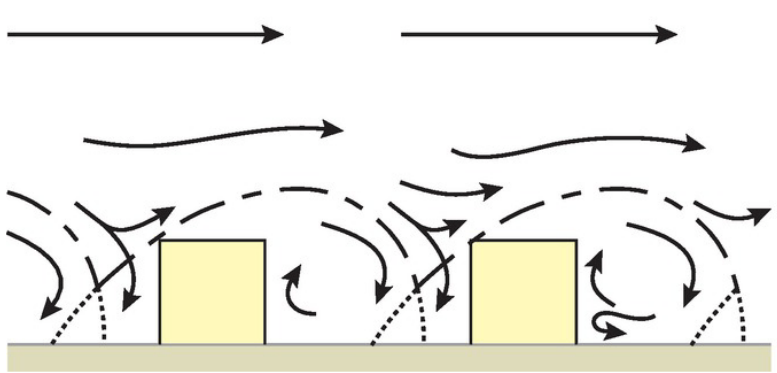
(Oke, 1988)

$AR = W_{sc}/h_{sc} > \sim 1.5$

(a) Isolated roughness flow

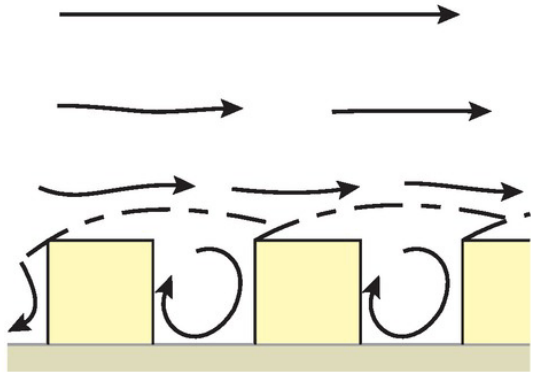


(b) Wake interference flow



$\sim 0.5–0.7 < AR < \sim 1.5$

(c) Skimming flow



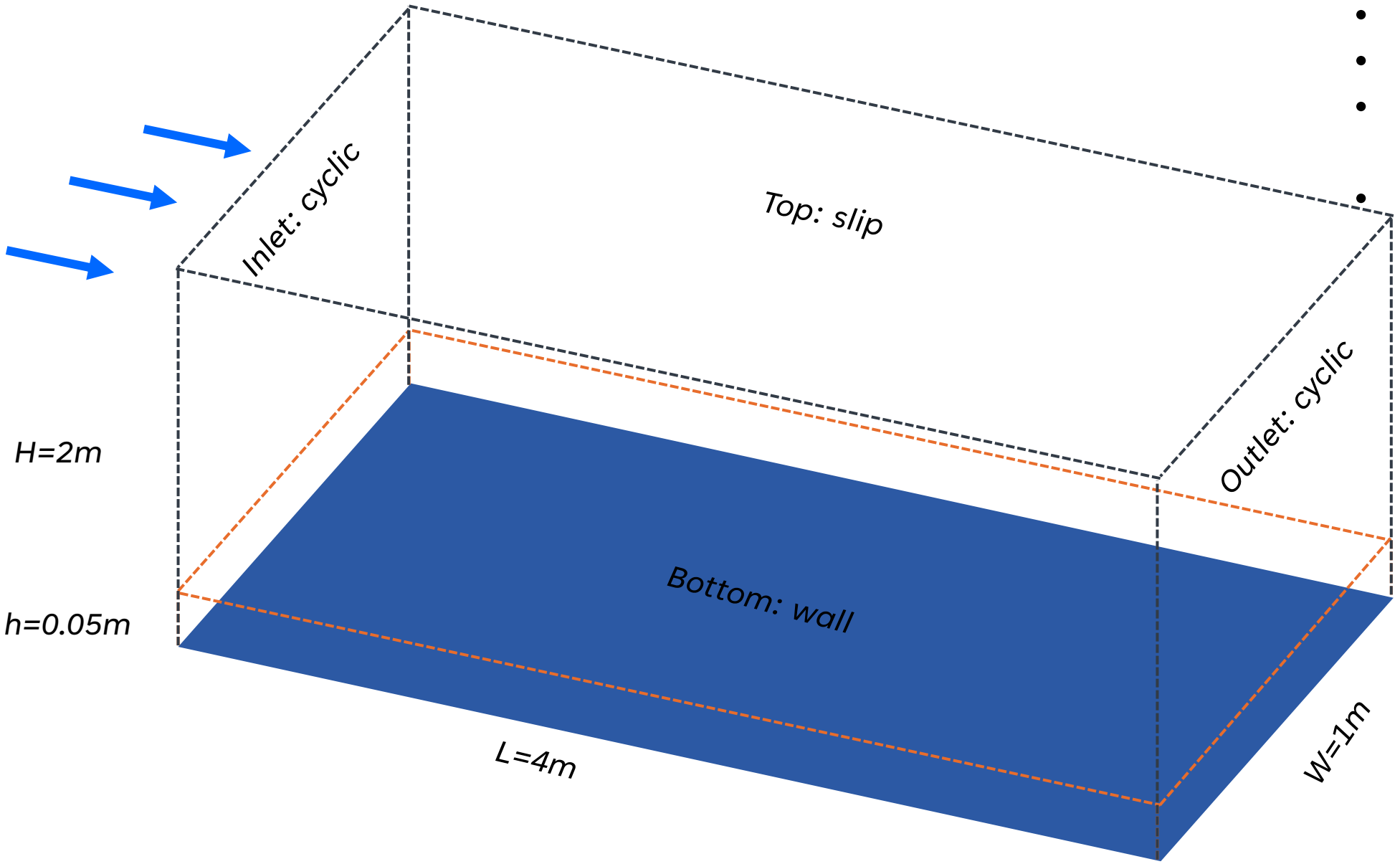
$AR < \sim 0.3–0.5$

Research Questions

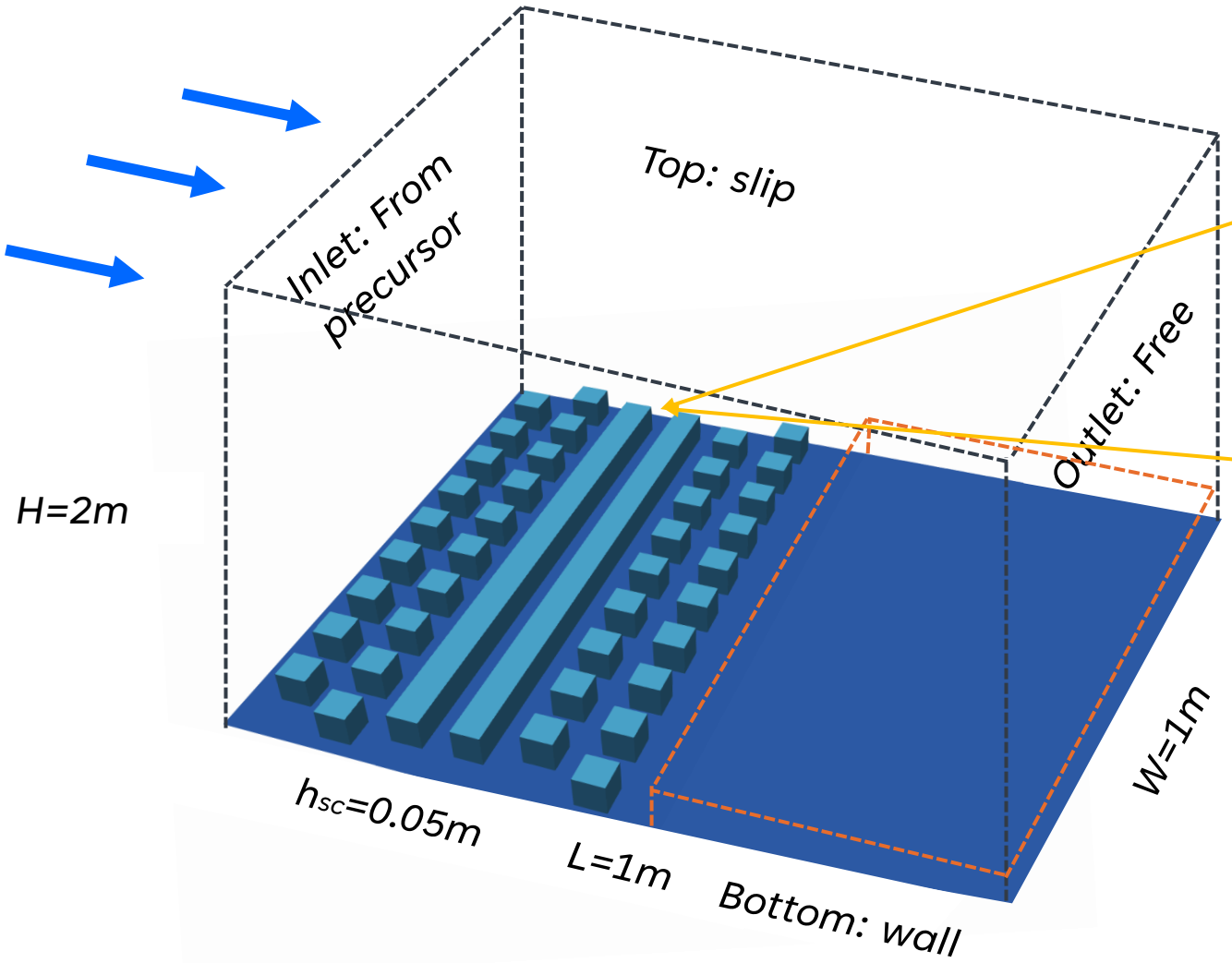
> Research Questions

1. Use drag porosity method reproduce a realistic urban boundary layer in OpenFOAM.
2. How do the turbulence statistics vary within street canyons of different aspect ratios in CFD simulations?

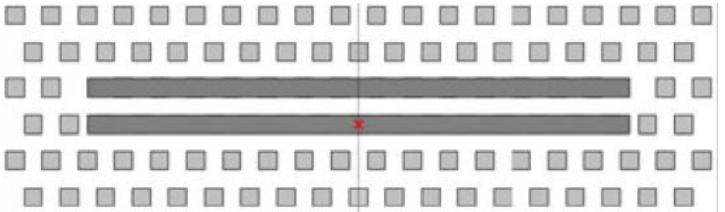
Simulation Set-ups



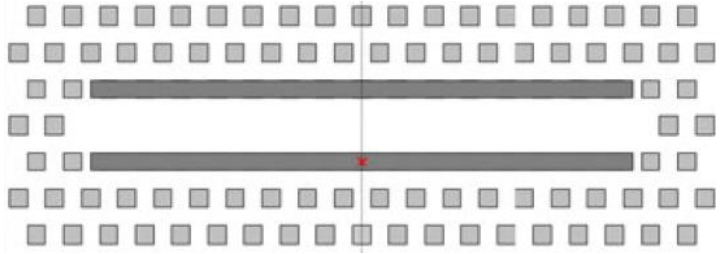
- OpenFOAM V2406
- Pimplefoam LES – TKE
- Output1: Fields U, k at outlet
- Output 2: Probes at L=2m, several different heights



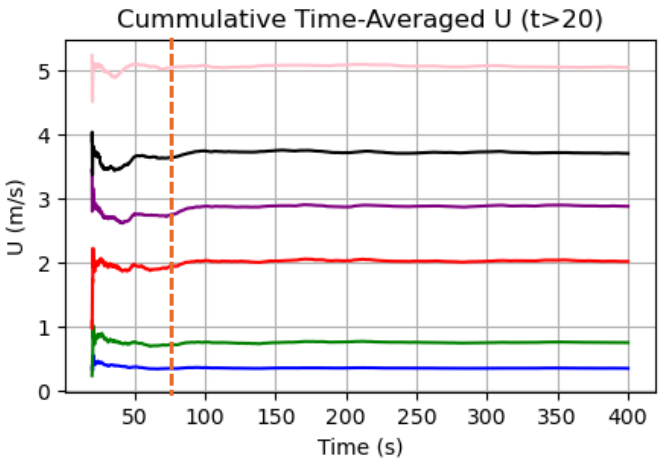
$AR = W_{sc}/h_{sc} = 1$

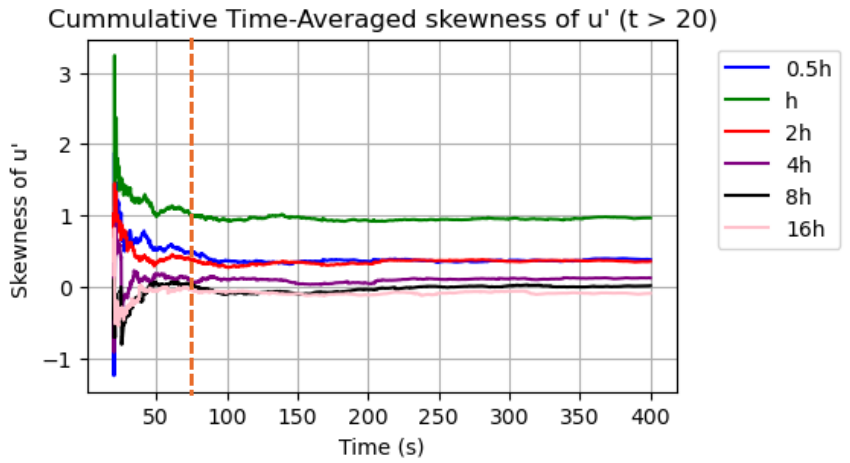


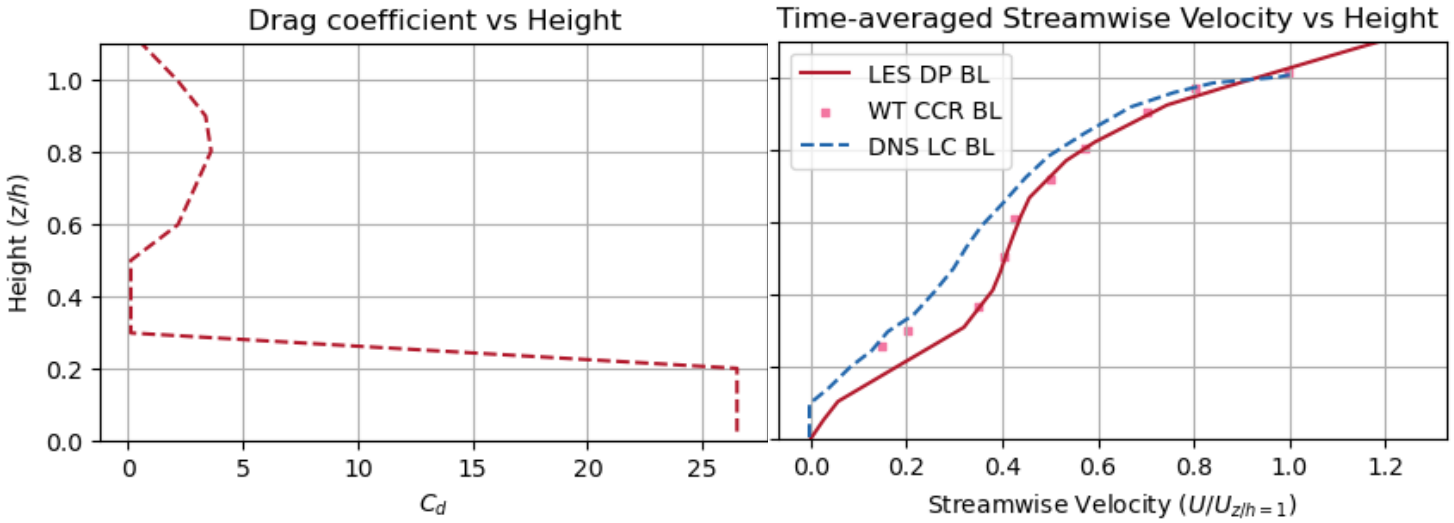
$AR = W_{sc}/h_{sc} = 3$



Results


Cummulative Time-Averaged U (t>20)
U (m/s)
Time (s)

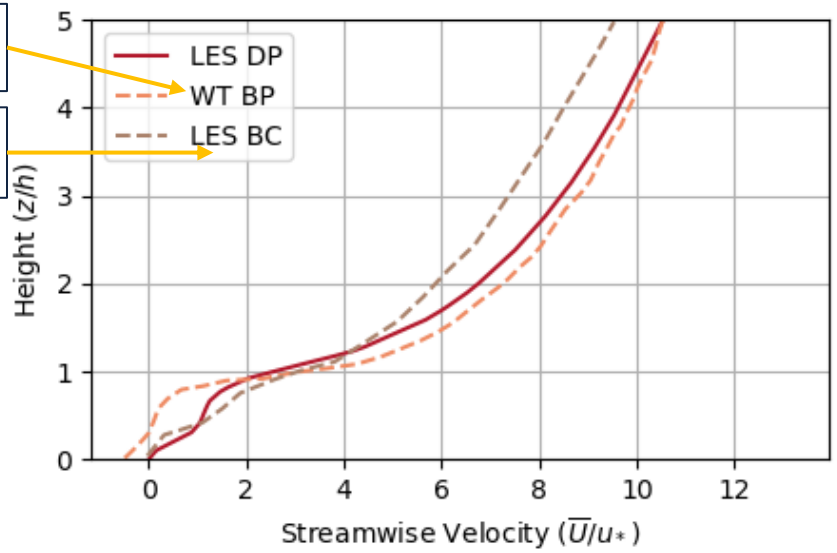

Cummulative Time-Averaged skewness of u' (t > 20)
Skewness of u'
Time (s)
0.5h
h
2h
4h
8h
16h

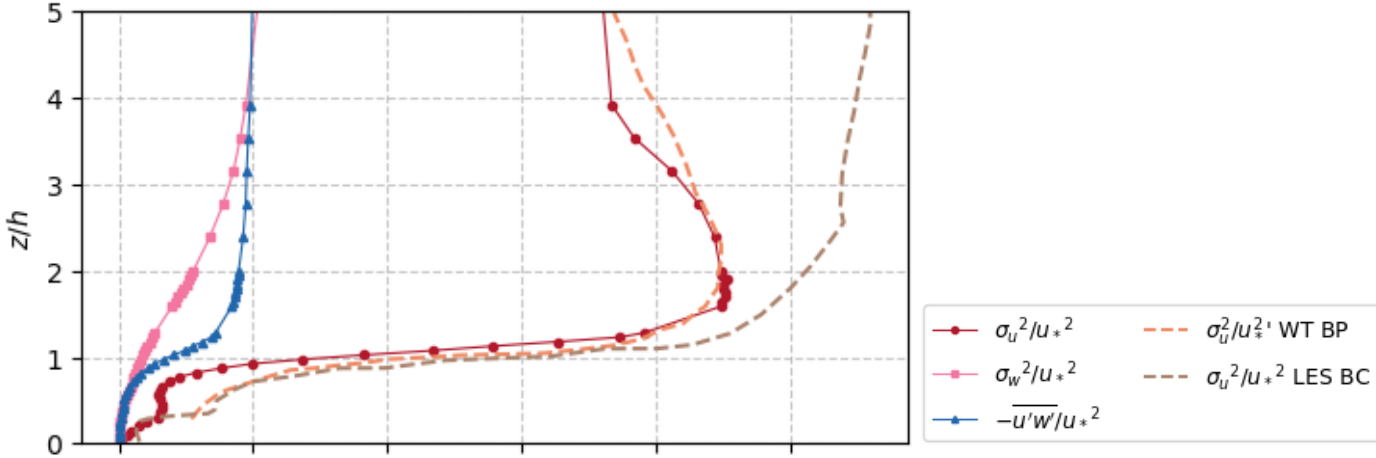

Drag coefficient vs Height
Height (z/h)
C_d
Time-averaged Streamwise Velocity vs Height
LES DP BL
WT CCR BL
DNS LC BL
Streamwise Velocity ($U/U_{z/h=1}$)

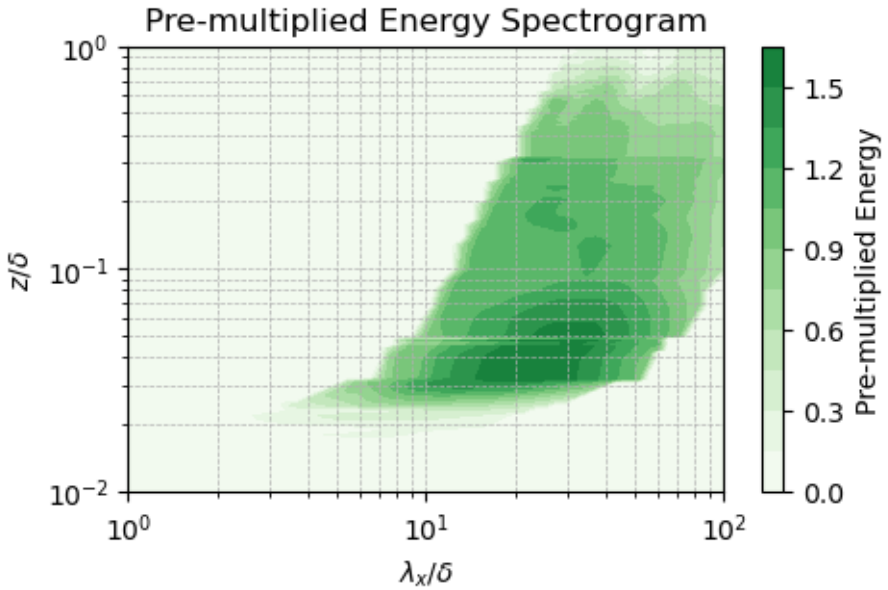
> Results > Precursor

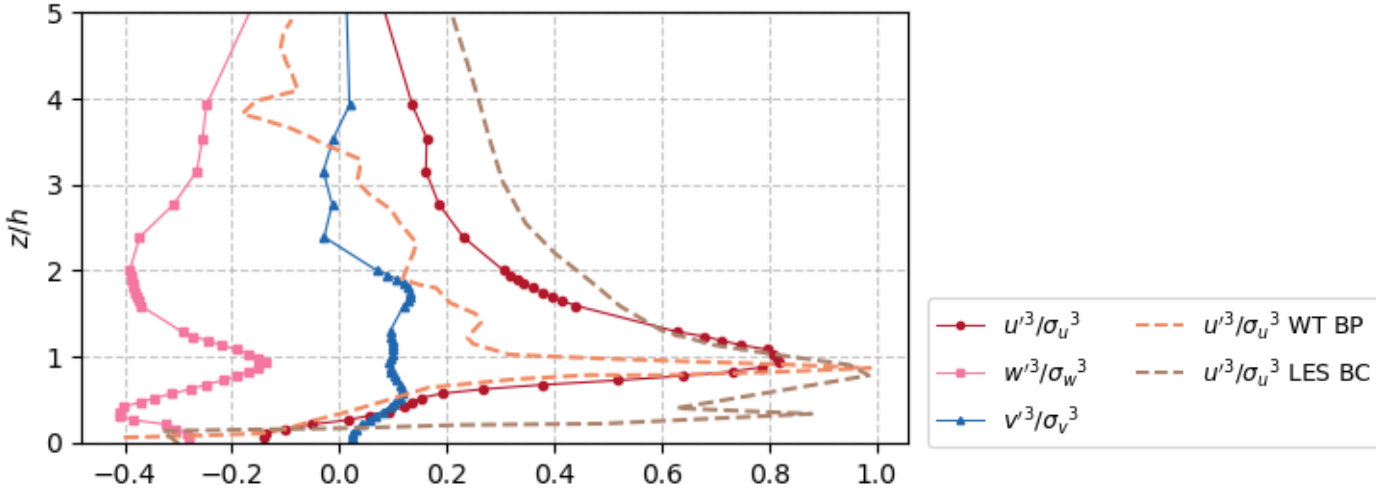
(Blackman and Perret, 2016, POF)

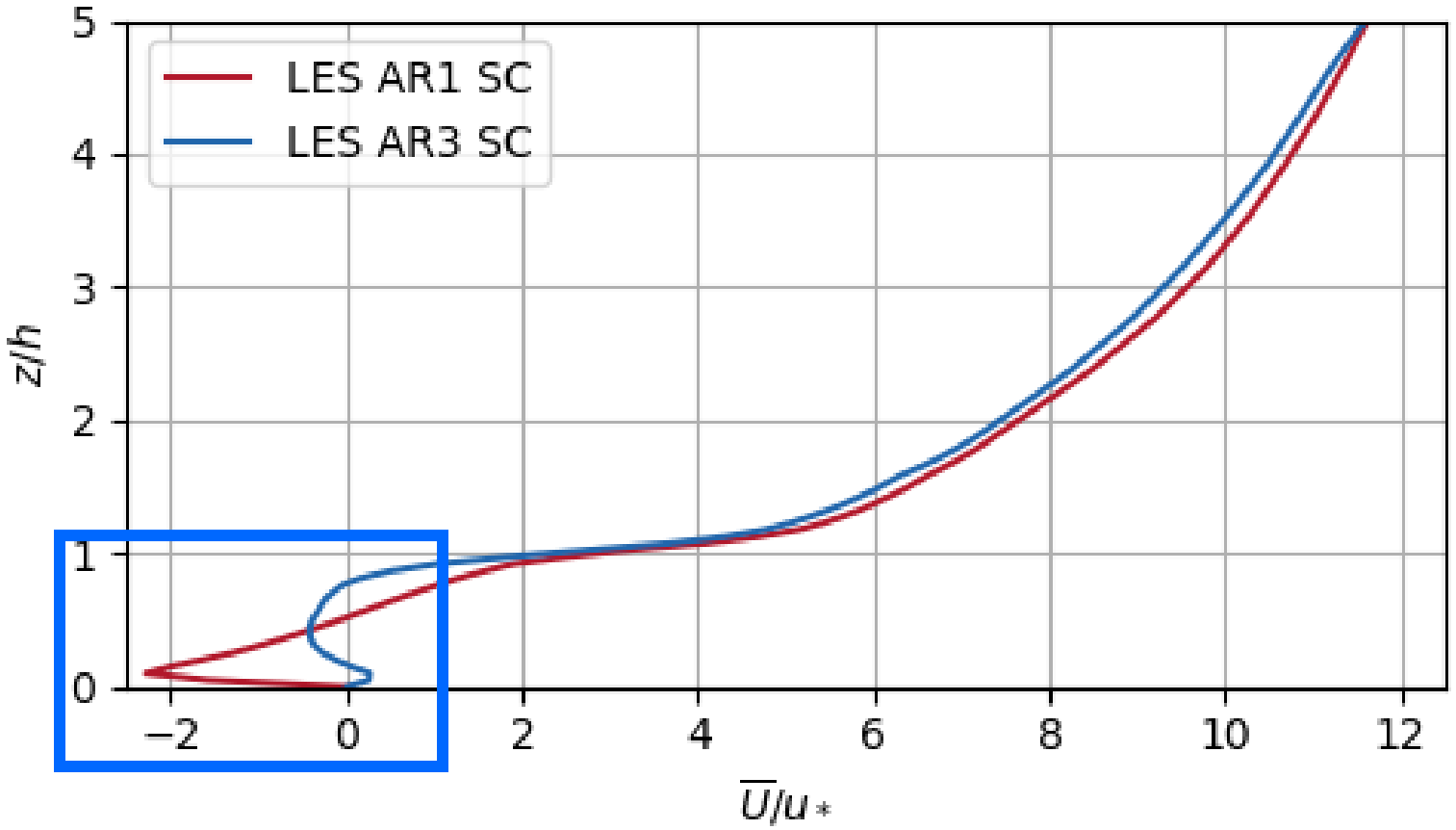
(Bucquet et al., 2023 JWEIA)

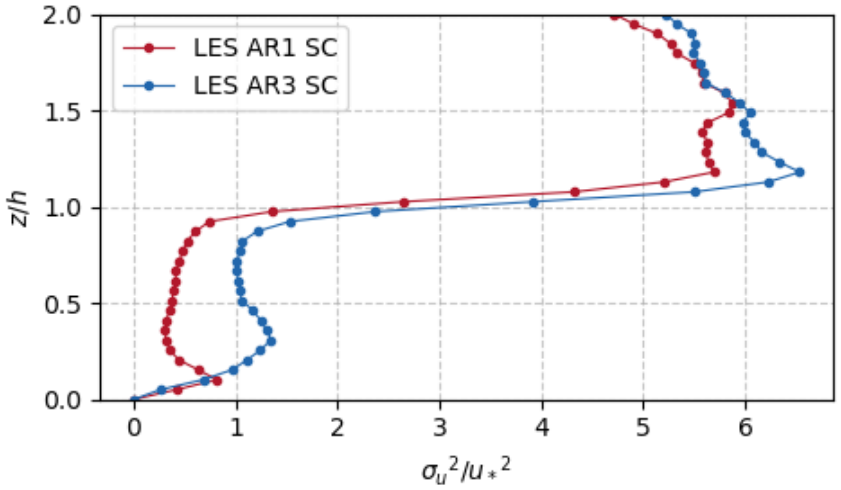


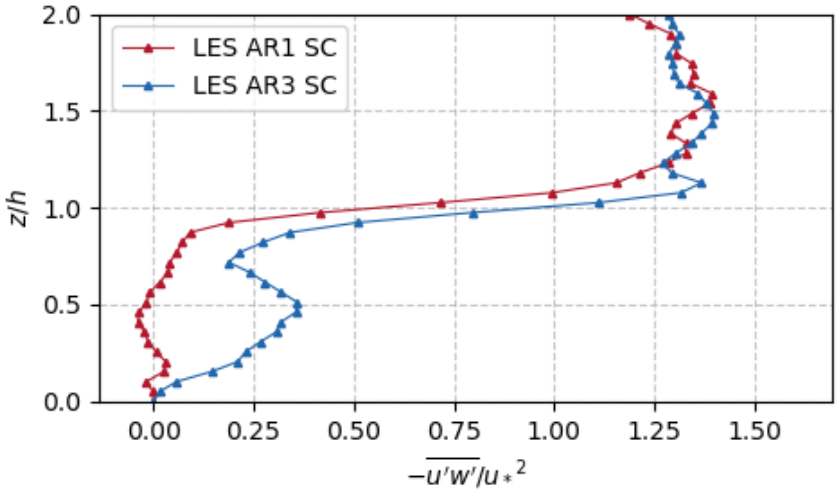


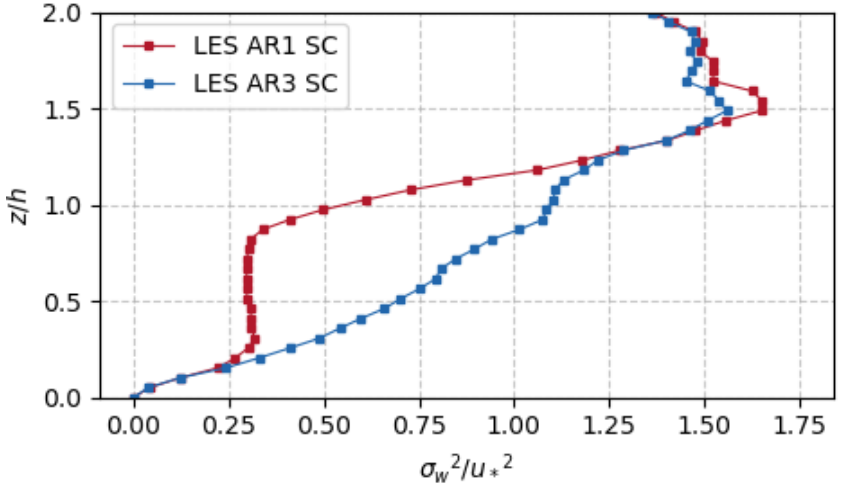


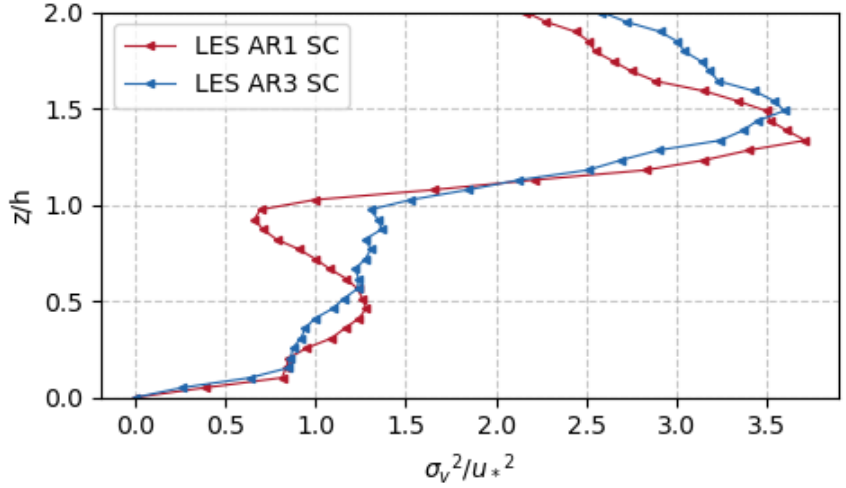


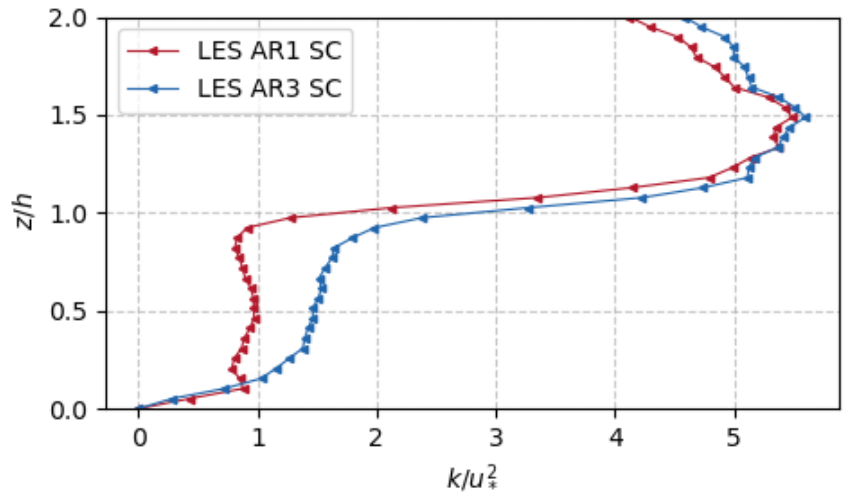

LES AR1 SC
LES AR3 SC
z/h
$\overline{U}/u_*$
0
1
2
3
4
5
−2
0
2
4
6
8
10
12


LES AR1 SC
LES AR3 SC
z/h
σ_u^2/u_*^2


LES AR1 SC
LES AR3 SC
z/h
$-\overline{u'w'}/u_*^2$


LES AR1 SC
LES AR3 SC
z/h
σ_w^2/u_*^2


LES AR1 SC
LES AR3 SC
z/h
σ_v^2/u_*^2


LES AR1 SC
LES AR3 SC
z/h
k/u_*^2

Summary

> Summary

1. The urban boundary layer is reproduced using the drag porosity method.
2. Street canyons of different regimes (isolated or wake interference) show very different turbulence behaviours at the centreline.
3. Impact of the street canyon aspect ratio diminishes above the shear layer.

Future Work

> Future Work

1. As still at a very early stage, simulations on street canyons should be refined, in near-wall regions and shear layers.
2. Wind tunnel data were gathered by other previous group members, and will be analyzed to compare with the simulation results.
3. Integrate tall buildings in the successor models to determine the interaction between the tall building wake and street canyon flows dynamically, and compare with my wind tunnel results with a morphological model.
4. I have developed a reduced-order model using the overlying boundary layer to predict the mass flux in the street canyon. However, it does not work with the presence of a tall building. In simulation, I believe I can get better understanding in tall building wake in 3D and then I can develop my model based on the interaction of the wake and the street canyon flow.

Thank you!

NSERC CRSNG

Western UNIVERSITY · CANADA



Dr. Eric Savory

AGENCE NATIONALE DE LA RECHERCHE ANR

CENTRALE NANTES



Dr. Laurent Perret